

Chapter 14. Transport Layer Security

14.1. Introduction

Transport Layer Security (TLS) is a cryptographic protocol designed to secure data transmission over computer networks and is widely used to ensure data privacy, integrity and authenticity. Matter TLS compatibility is intended to provide a mechanism for Application layers to use TLS to communicate securely with devices or servers outside the Matter [Fabric](#).

Since TLS employs digital certificates to authenticate both the identity of servers and clients, this chapter discusses the provisioning of nodes with X.509 digital certificates, as well as the provisioning of nodes with other TLS endpoint information needed to facilitate TLS communication.

Nodes MAY choose to use TLS in other, non-Matter, contexts. Therefore in this chapter all references to Nodes supporting or using TLS, and the mechanisms defined to manage TLS connections, pertain only to the use of the TLS within Matter features that depend on it.

14.2. Common Conventions

The following conventions are used.

14.2.1. Transport Layer Security (TLS)

Nodes supporting TLS SHALL implement Versions 1.3 (as defined in [RFC 8446](#)) or later, and SHALL NOT support any earlier version. Matter's use of TLS SHALL support only key exchange/cipher suites which provide Perfect Forward Security.

Since Matter TLS connections by their nature connect to services outside the Matter Fabric, TLS connections MAY utilize IPv4 transport. Ecosystems implementing Matter linked TLS services SHOULD therefore ensure their TLS services have both IPv6 and IPv4 connectivity.

14.2.2. TLS Certificates

TLS certificates used within this chapter SHALL be based on X.509v3-compliant certificates as defined in [RFC 5280](#). TLS certificates used in this chapter SHALL be encoded using the Distinguished Encoding Rules (DER) format as defined in [X.690](#).

14.2.2.1. X.509 Certificate Fingerprint

An X.509 certificate fingerprint is created by applying a cryptographic hash function to the X.509 certificate's binary data, enabling easy comparison and verification. X.509 certificate fingerprints used in this chapter SHALL be calculated by using the SHA-256 hash algorithm as defined in [RFC 6234](#), using the DER (binary) encoding format of the certificate.

14.2.2.2. Certificate Authority (CA)

A Certificate Authority (CA) is a trusted entity that issues digital certificates. These digital certificates are used in Public Key Infrastructure (PKI) to facilitate secure communication.